XRPL Validator — Daily Engineering Report · 2026-10-29

Period: 2026-10-29 00:00 to 24:00 US Eastern (2026-10-29 04:00 to 2026-10-30 04:00 UTC); ledgers #107,972,242 to #107,993,767. Every count from the ledger and the validator's logs stops at #107,993,767, closed 03:59:52 UTC. Generated 2026-10-30 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Degraded (all causes resolved or fixed) — coverage outside planned maintenance 99.814% (objective 99.9%): 40 ledgers unsigned.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,486 of the 21,526 ledgers the network closed (99.814%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 40 outside maintenance (1 incident).
- **Native engine:** 1 engine receipt: #107,989,363 — fixed: commit b57328d (2026-10-30).
- **Changes:** 0 deployments, 0 commits merged into main.
- **Independence program:** 1 of 30 qualifying days (snapshot at 04:20:46 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,526
Validations issued	21,486
Coverage (validations / ledgers closed)	99.814%
Validations matching the network's validated ledger hash	21,486 of 21,486
Unsigned during planned restarts	0
Unsigned outside maintenance	40
Coverage outside planned maintenance (objective 99.9%)	99.814%
Restarts	0
Basis: our own ledger hash	21,486
External agreement, data.xrpl.org (UTC day 2026-10-29)	0.99806 (40 missed of 20,667)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,972,242–#107,993,767), by step:

Step	Ledgers with a receipt
Step 1 · native engine	1
Step 2 · records	1
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	1
Writer	1

Live counters, snapshot at 04:20:46 UTC (since the last restart, 2026-10-28 22:52 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	26,080 ledgers, 5,281,276 transactions	26,079 fully matched; results 0, bytes 1, keys 0/0
2 · transaction records	26,373 ledgers, 5,386,906 records	5,386,905 exact; differ 1, tree 1
3 · transaction order	26,413 ledgers (mode: on)	26,413 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	26,461 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	26,458 ledgers judged	26,458 agreed; differ 0, short of quorum 0
5 · our own ledger hash	26,460 ledgers	26,372 own; differ 1, partial 87
Writer	every ledger written	mismatch 1, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	23:04 UTC	#107,989,363	Step 1 · native engine, Step 2 · records, Step 5 · own hash, Writer	SignerListSet tx 0A9A6F552C51: AccountRoot 13F1A95D (modified on the network): FinalFields.OwnerCount ours=13 theirs=10	fixed: commit b57328d (2026-10-30); awaiting merge and deployment; see INC-01

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-29-01, INC-01 is INC-2026-10-29-01.

Incidents

INC-01 · 40 ledgers unsigned

Window	23:04:30 UTC to 23:07:00 UTC
Ledgers	#107,989,363–#107,989,402 (40)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-01
Cause	#107989363 0A9A6F552C51 (SignerListSet, SignerQuorum 0) deleted a 2-entry list made before MultiSignReserve: no lsfOneOwnerCount, so rippled's removeSignersFromLedger gives back 2 + entries. Mainnet's OwnerCount went 14 -> 10; ours went 14 -> 13, because the delete branch always gave back one. The replacement path and AccountDelete already had the legacy rule; the delete branch now uses the same one, through owner_count_add.
Action	commit b57328d — fix(ledger): deleting a legacy SignerList gives back 2 + its entries; awaiting deployment

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **1 of 30** (snapshot at 04:20:46 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-29 — tracked #107,968,634–#107,990,190	Value
Validations	21,517 (21,517 own, 0 borrowed)
Coverage of the tracked ledgers	99.81%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,526 (#107,972,242–#107,993,767)
Mean close interval	4.01 s
Longest close interval	10 s (before #107,993,418)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Receipt R-01 (#107,989,363): fixed: commit b57328d (2026-10-30); awaiting merge and deployment; see INC-01.
- Committed, awaiting merge and deployment: `3b4b64c` feat(writer): a hash failure patches the mirror in place; the breaker is an alarm only.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.